Anchored Causal Action-Contrast Transfer for Cross-Network Traffic Signal Control

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Abstract

Controllers transferred between traffic networks must rank actions under simultaneous changes in topology, demand and signal programs. Dense simulator-residual models can predict source transitions accurately while carrying action-common domain error into the target decision. We introduce anchored Causal-Factored Cross-City Model Transfer (CFCMT), a target-adapted action-ranking method that learns only matched counterfactual contrasts. For every restored simulator state, a rigid model predicts candidate-minus-reference cost from a declared local parent set and an all-action model predicts antisymmetric pairwise cost differences. A seed-blocked target selector chooses a constant or confidence-capped interpolation between the two, after which one model is refitted on all available adaptation groups. The pressure reference defines the contrast coordinate but is not a deployment fallback.

Development used 18 SUMO networks from seven city groups. On an evaluation seed excluded from fitting and selection, leave-one-city-out normalized action regret decreased from 0.241 to 0.218 (9.84%; six of seven city groups improved). We then froze the method before collecting labels from two external city-derived networks. Using uncalibrated target simulation from two adaptation seeds, the untouched-seed city-macro regret was 0.269, compared with 0.292 for the rigid anchor and 0.310 for a target-label-only comparator. In 56 matched 3,600-s closed-loop rollouts, CFCMT reduced all-departed waiting time from 159.15 to 149.17 s relative to an H2O^{+}-style dense residual baseline, a 6.27% gain with a descriptive paired interval of [2.01, 17.19] s. A preregistered source-contribution test then paired the frozen CFCMT and target-label-only controllers on 64 new seeds and 512 rollouts. CFCMT improved Jinan by 3.04% but regressed Los Angeles by 7.87%; the equal-city relative difference was +4.25% (95% interval $[-0.77, 13.21]$%), so the joint confirmation was rejected. MaxPressure and phase pressure also remained stronger in the original matrix. The evidence supports matched action contrast over dense residual transfer on these benchmarks, while showing that source data can induce target-network-specific negative transfer; it does not establish zero-shot transfer, field causal identification or universal benefit from cross-city data.

1 Introduction

Traffic-signal controllers are commonly trained and tested within one fixed network. New random seeds change arrivals, but topology, route choice, phase programs and feature semantics remain largely unchanged. Cross-network deployment changes all of them at once. A policy or world model can therefore fit source transitions while misranking actions in a target network, which is the quantity that ultimately determines control performance.

Two established approaches expose the central trade-off. MaxPressure and phase-pressure controllers use local queue imbalance and transfer without a city-indexed policy, with strong throughput and stability motivation (Varaiya, 2013). Pressure structure also underlies learned controllers including PressLight, FRAP and MPLight (Wei et al., 2019a; Zheng et al., 2019; Chen et al., 2020). These rules are difficult baselines because local pressure is already close to the control sufficient statistic in many signal networks. Conversely, simulator-based offline-to-online reinforcement learning can use source data to correct dynamics mismatch (Niu et al., 2022, 2025), but an unrestricted residual may absorb network identity, demand scale and phase-specific associations that do not survive a city shift.

Prior transfer-oriented signal control spans materially different information budgets. MetaLight adapts a meta-learned initialization with target experience, whereas GESA and X-Light construct scenario-normalized or cross-city representations for policy generalization (Zang et al., 2020; Jiang et al., 2024a,b). CrossLight explicitly combines source-city offline pretraining with target-city online fine-tuning (Sun et al., 2024). Sim-to-real work such as UGAT instead transforms actions to compensate for a simulator dynamics gap (Da et al., 2023). These methods establish that topology normalization, meta-adaptation and dynamics-gap correction are each useful, but their zero-shot, online-interaction and sim-to-real protocols answer different questions. Our setting is target offline adaptation: target interactions are counterfactual simulator labels collected before deployment, and no target closed-loop reward is used for model selection.

Mechanism-factored world models provide an intermediate design. They restrict prediction to stable parent sets and can adapt mechanism state separately, consistent with the modularity motivation of causal transfer (Bengio et al., 2020; Feng et al., 2022; Schölkopf et al., 2021). Our experiments revealed a sharper requirement for traffic control: accurate decomposition of absolute dynamics does not guarantee accurate ordering of feasible phases. The final method therefore retains structural parent restrictions but changes the estimand from next-state prediction to within-state action contrast.

We call the resulting method anchored Causal-Factored Cross-City Model Transfer (CFCMT). At a simulator snapshot, every feasible phase is executed from the same restored state. Candidate cost is expressed relative to one source-selected pressure reference and normalized by the action range of that matched group. A low-capacity rigid anchor estimates candidate-versus-reference cost from declared queue, occupancy, pressure and transition parents. A second model compares every unordered phase pair and is antisymmetrized by construction. The target score is

$$S(s,a) = A(s,a) + \alpha(s)\{P(s,a) - A(s,a)\}, \tag{1}$$

where the interpolation is selected out of fold by holding out one complete target adaptation seed. Constant interpolation permits aggressive correction when it is stable; a local confidence-cap limits how far the pairwise model can perturb the anchor when they disagree.

The information budget is central to the claim. CFCMT is not evaluated as zero-shot transfer in the final external experiment. It uses target-network files, online target state and counterfactual transition labels generated by an uncalibrated target SUMO model. Those labels constitute target offline adaptation even though no target simulator parameter is fitted to field data. The method is therefore calibration-light rather than target-data-free. Adaptation seeds, the offline confirmation seed and closed-loop seeds are disjoint, and the evaluation cache did not exist when the target model and selector were frozen.

The empirical design has four levels. First, 18 source networks from seven city groups support leave-one-city-out development with an excluded evaluation seed. Second, two external city-derived networks, Los Angeles and Jinan, test the frozen procedure using all valid target adaptation groups and a new offline seed. Third, 56 matched 3,600-s closed-loop rollouts compare CFCMT with fixed time, simulator-only model predictive control, a rigid causal anchor, an H2O$^+$-style dense residual and two classical pressure policies. All policies share target networks, feasible green phases, demand seeds, signal-transition execution and metric populations. Fourth, a target-label-only model already frozen in the same baseline family is compared with CFCMT on 64 newly generated seeds, four target scenarios and 512 rollouts. This final stage directly tests whether source-labelled counterfactuals improve closed-loop control after the target adaptation budget is held fixed. Official RESCO and LibSignal reinforcement learning runs are retained only as native-protocol integration checks because their state, reward, action timing and training budgets are not commensurate with this experiment (Ault and Sharon, 2021; Mei et al., 2024). The study consequently does not present native CrossLight, X-Light or GESA scores as if they were paired baselines; a same-protocol reproduction of those systems remains outside the evidence reported here.

This study makes four contributions. First, it formulates cross-network signal adaptation around matched action contrasts and proves cancellation of action-common domain bias. Second, it combines a parent-restricted anchor with an antisymmetric all-action correction and derives exact ranking and bounded perturbation properties. Third, it implements an auditable seed-blocked selection and full-refit protocol that prevents out-of-fold or evaluation models from entering deployment. Fourth, it separates structured-model gain from source-data gain. CFCMT improves the dense residual, but a fresh target-label-only comparison reveals positive source contribution in Jinan and negative transfer in Los Angeles, rejecting a general source-benefit claim. Classical pressure control remains the closed-loop ceiling, and a separate 64-seed execution-layer challenge does not close that gap. The contribution is consequently an auditable action-ranking framework and a falsifiable account of where cross-network transfer fails, not a claim that complexity or source volume alone should replace a strong traffic-control rule.

2 Results

2.1 A frozen action-contrast transfer protocol

The final experiment evaluates target offline adaptation rather than zero-shot policy transfer. Source counterfactuals and target counterfactuals are both generated from matched restored SUMO states, but target roles are separated by seed (Figure 1). Seeds 5057 and 6067 are the only target adaptation seeds. Seed 8171 is collected after the joint method freeze and is used once for offline action-regret confirmation. Seeds 8081 and 9091 are used only for closed-loop policy rollouts. Out-of-fold models select the blend but are never deployed; after selection, each external city receives one refit on the union of all its valid adaptation groups.

The source bank contains 18 networks in seven city groups: Atlanta, Cologne, Hangzhou, Ingolstadt, New York, two RESCO synthetic grids and Salt Lake City. Together they contain 317 controlled traffic lights and 6,076 non-internal lanes. The external set contains a four-signal Los Angeles corridor and a 12-signal Jinan grid evaluated under three demand files (Figure 2; detailed counts in Table 6). City groups, rather than network files, are the development transfer unit, so another Cologne, Hangzhou or Ingolstadt subnetwork cannot enter the training fold when that city is held out.

2.2 Seed-held-out development selected a non-anchor correction

The method was not accepted because it fit its development seed. Models and the city-level selector were fitted using source seeds 2027 and 3037, then evaluated on seed 4047. In leave-one-city-out evaluation, the rigid anchor had city-macro normalized action regret 0.2414 and the selected anchored model had regret 0.2177, a 9.84% reduction (Table 1). Six of seven held-out city groups improved. Hangzhou was the sole regression, with an absolute increase of 0.00388, below the frozen 0.01 tolerance.

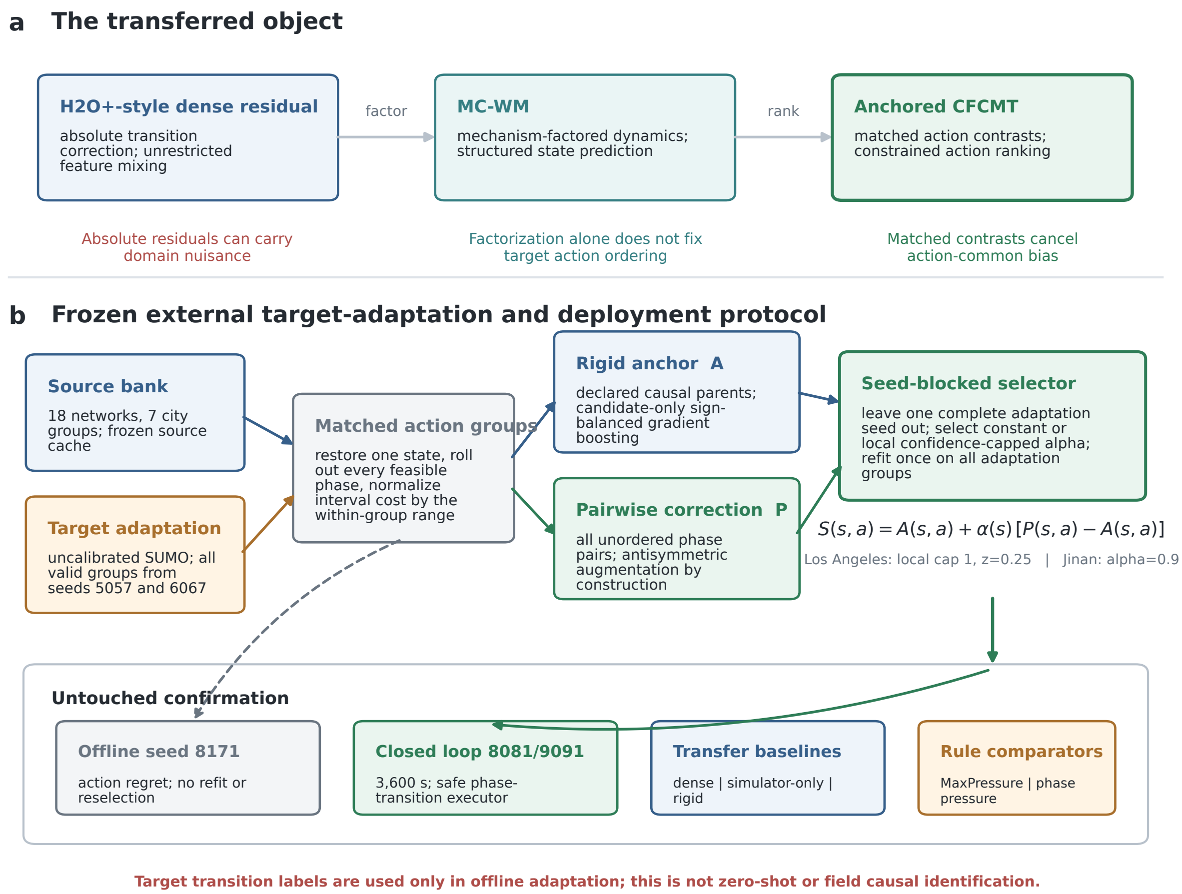

Figure 1: Transferred object and frozen evaluation protocol. **a**, The method lineage contrasts absolute dense-residual prediction, factored MC-WM dynamics and the final matched-ranking estimand. The same-protocol dense baseline in panel **b** is deliberately strengthened with the matched contrast data and target refit budget used by CFCMT. **b**, A rigid parent-restricted anchor and an antisymmetric all-pair correction are combined by a seed-blocked target selector. The pressure reference defines the contrast origin; MaxPressure and phase pressure are independent comparators, not fallback actions. Target counterfactual labels make the final setting target offline adaptation rather than zero-shot transfer.

Table 1: Seed-held-out development confirmation. Regret is candidate cost minus the matched-group optimum, normalized by the within-group action range; lower is better.

Development result	Anchor regret	Selected regret	Relative gain
Seven-city leave-one-city-out	0.241	0.218	9.84%

Earlier variants failed their gates. A global pairwise correction regressed on the added Salt Lake domain, the first anchored selector exceeded its worst-city tolerance, and a cross-fitted selector did not achieve the required city-level stability. These failures were retained and the successful seed-held-out protocol was frozen before external labels were generated (Table 7).

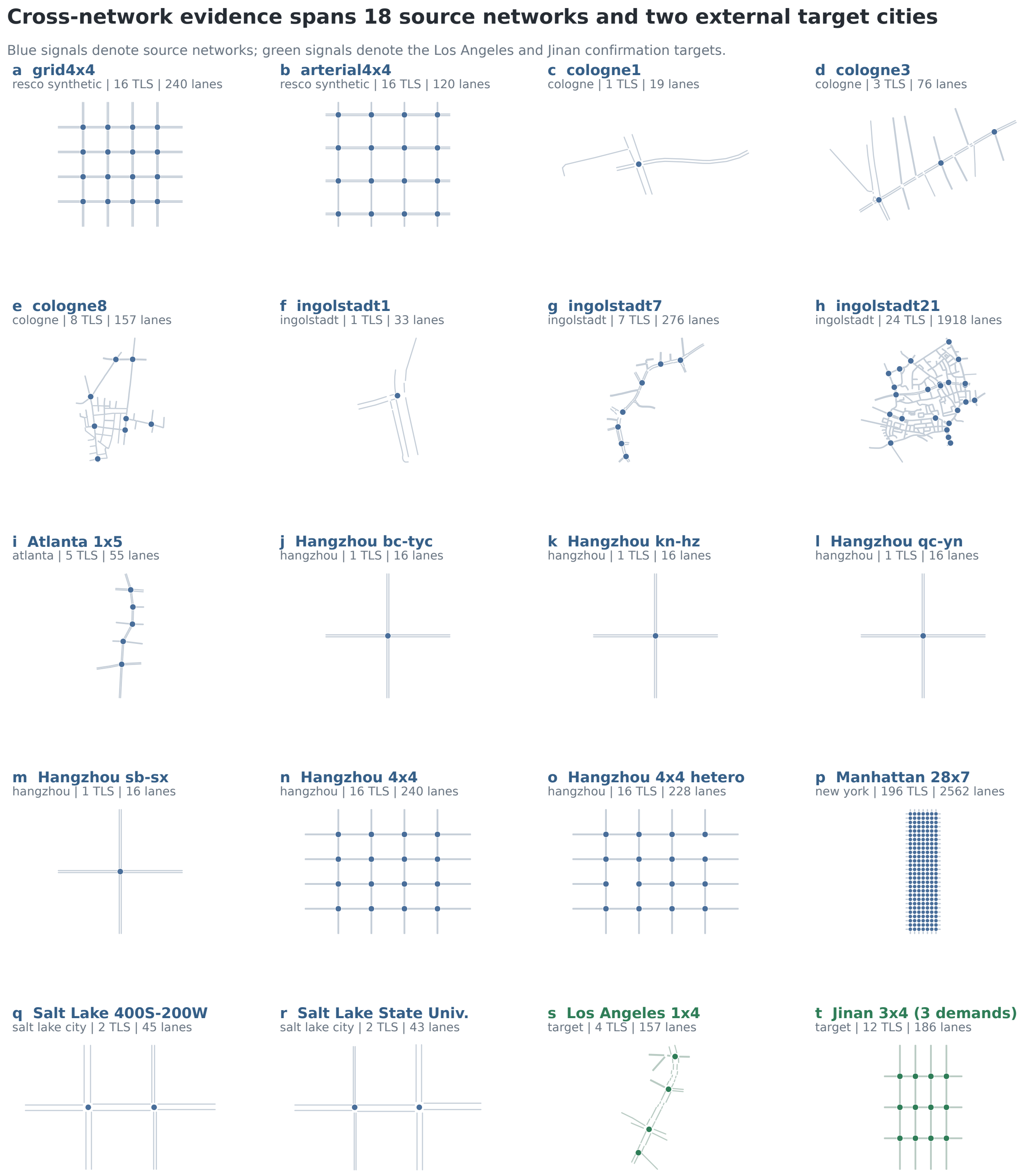

Figure 2: Network shift in development and external confirmation. Lines show non-internal SUMO lanes and points show controlled traffic lights. Blue panels are the 18 source networks grouped into seven development cities; green panels are external target geometries. The Jinan geometry is paired with three demand files. The plate documents topology and action-set variation; it does not imply that the converted networks are calibrated city-wide digital twins.

Table 2: External seed-8171 offline confirmation after city-specific method freeze. Target only uses the same causal action-ranking family and target groups but zero deployed source weight. Adaptation groups are target counterfactuals from seeds 5057 and 6067; held-out groups are never used for fitting or selection. Gains in the table compare CFCMT with target only.

City	Adapt. groups	Held-out groups	Anchor	Target only	CFCMT	CFCMT vs target only	Selected blend
Los Angeles	165	79	0.148	0.217	0.145	33.24%	local_cap_1_z_0p25
Jinan	911	457	0.436	0.404	0.392	2.85%	constant_alpha_0p9
External macro	–	2 cities	0.292	0.310	0.269	13.48%	city-specific

2.3 External offline scores favoured source-plus-target fitting

The external selector used every valid adaptation group: 165 in Los Angeles and 911 across the three Jinan demands. Holding out each complete adaptation seed produced two selection folds per city. Los Angeles selected a local blend with disagreement cap 1 and confidence multiplier 0.25; Jinan selected a constant correction weight of 0.9. The deployment artifacts were then refit once on all adaptation groups and frozen before seed 8171 was collected.

On seed 8171, Los Angeles regret decreased from 0.1482 to 0.1450 (2.18%) and optimal-action accuracy increased from 0.785 to 0.810. Jinan regret decreased from 0.4361 to 0.3921 (10.10%) and optimal-action accuracy increased from 0.296 to 0.342 (Table 2 and figure 3a). Equal weighting of the two cities gives a regret reduction from 0.2922 to 0.2685, or 8.09%, and both city point estimates improve.

The freeze also contained a target-label-only comparator from the same causal action-advantage family. It used the same feature semantics and target adaptation groups, but its target prediction weight was numerically one, so source-labelled groups did not contribute to its deployed score. Its seed-8171 regret was 0.2172 in Los Angeles and 0.4036 in Jinan. CFCMT therefore reduced the equal-city point estimate from 0.3104 to 0.2685 (13.48%). The paired action-group improvement was 0.07219 in Los Angeles (95% interval $[0.01671, 0.13512]$, two-sided bootstrap probability value 0.011) and 0.01150 in Jinan (interval $[-0.02250, 0.04524]$, probability value 0.5196). Offline source contribution was therefore supported in Los Angeles but inconclusive in Jinan, despite both point estimates favouring CFCMT.

The uncertainty differs sharply by city. A 10,000-replicate paired group bootstrap gives a Los Angeles improvement interval of $[-0.0675, 0.0729]$ with a two-sided bootstrap probability value of 0.9396, whereas Jinan gives $[0.00469, 0.08293]$ and 0.0274. The Los Angeles result is therefore directional, not independently conclusive. The external gate is a predeclared engineering confirmation (at least 2% city-macro improvement, both cities improving and no city regression), not a population-level test over cities.

2.4 Closed-loop transfer improved dense residual control

The formal closed-loop matrix contains four target scenarios, two seeds and seven policies, yielding 56 completed 3,600-s rollouts. The primary metric is mean waiting over every departed vehicle. SUMO `write-unfinished` records accrue waiting to the fixed horizon, so unfinished vehicles remain in the population. Completed-only waiting, system vehicle-hours and completion are reported beside it to expose censoring and throughput trade-offs.

Anchored CFCMT achieved 149.17 s all-departed waiting, compared with 159.15 s for the H2O^{+}-style dense residual, 156.71 s for simulator-only control and 152.91 s for the rigid anchor (Table 3). Relative to the dense residual, the reduction was 9.98 s or 6.27%; both Los Angeles (8.46 s) and Jinan (11.50 s) improved. The descriptive paired hierarchical bootstrap interval was

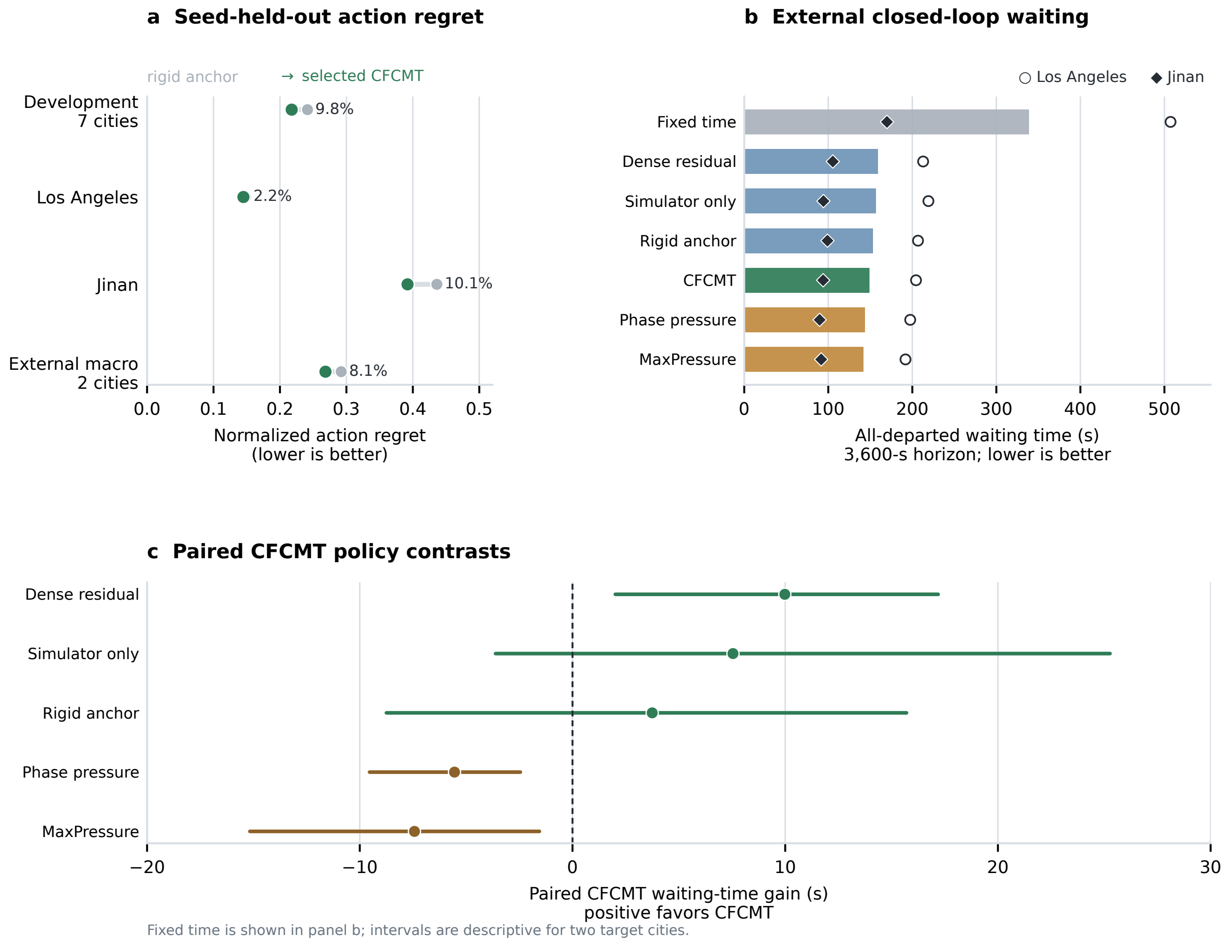

Figure 3: Independent offline and closed-loop confirmation. **a**, seed-held-out normalized action regret for development and external targets. Green points show the selected CFCMT score and grey points the rigid anchor. **b**, all-departed waiting time by policy in Los Angeles (circles) and Jinan (diamonds); bars show the equal-city macro. **c**, paired CFCMT waiting-time gains relative to the non-fixed comparators; fixed time remains visible in panel **b** and Table 3. Positive values favour CFCMT. Intervals are descriptive paired hierarchical bootstrap intervals because there are only two independent target cities.

$[2.01, 17.19]$ s and 98.43% of replicates favoured CFCMT. Completed-only waiting was also lower (133.06 versus 147.47 s), system vehicle-hours decreased from 618.15 to 594.14, and completion increased from 0.870 to 0.889.

The point-estimate gains over simulator-only and rigid-anchor control were 7.54 s (4.81%) and 3.75 s (2.45%), respectively, but their intervals crossed zero (Table 4). Thus the closed-loop evidence is strongest against dense residual transfer. Improvement over the anchor is supported clearly by offline action regret and only directionally by these two closed-loop cities; the result does not justify attributing every policy gain to the pairwise correction.

Table 3: External 3,600-s closed-loop city-macro results. Scenarios are weighted equally within each city and the two cities are weighted equally. Waiting times are seconds; system burden is vehicle-hours. “Incidents” is the equal-city mean of unique SUMO junction-collision incidents.

Policy	All-departed wait	Completed wait	System veh-h	Unfinished	Completion	Incidents
Fixed time	338.58	308.72	885.18	0.133	0.661	2.00
Dense residual	159.15	147.47	618.15	0.066	0.870	0.25
Simulator only	156.71	138.68	619.66	0.069	0.877	0.00
Rigid anchor	152.91	138.40	606.30	0.072	0.882	0.50
Anchored CFCMT	149.17	133.06	594.14	0.067	0.889	0.00
Phase pressure	143.62	125.32	583.05	0.068	0.898	0.25
MaxPressure	141.74	135.42	580.29	0.065	0.896	0.25

Table 4: Paired CFCMT policy contrasts for all-departed waiting. Positive gains favour CFCMT. Intervals are descriptive, not population-level city inference.

Baseline	CFCMT gain (s)	Relative gain	Descriptive 95% interval	LA gain	Jinan gain
Fixed time	189.42	55.94%	[67.11, 308.32]	302.90	75.93
Dense residual	9.98	6.27%	[2.01, 17.19]	8.46	11.50
Simulator only	7.54	4.81%	[-3.62, 25.27]	14.75	0.34
Rigid anchor	3.75	2.45%	[-8.75, 15.70]	2.36	5.14
Phase pressure	-5.55	-3.86%	[-9.54, -2.45]	-6.91	-4.19
MaxPressure	-7.43	-5.24%	[-15.17, -1.56]	-12.53	-2.33

2.5 Fresh target-only comparison exposed network-dependent negative transfer

The seven-policy v43 matrix did not deploy its frozen target-label-only baseline. We first added that comparator to the same eight scenario–seed cells after seeing the main outcomes. This post-hoc v90 ablation left both models unchanged: CFCMT had 149.17 s city-macro waiting and target only had 149.04 s, a 0.12-s disadvantage. Los Angeles favoured target only by 2.45 s, whereas Jinan favoured CFCMT by 2.20 s. Because the comparator was added after the original result, this matrix is descriptive rather than confirmatory.

We then froze a v91 protocol before generating 64 further simulator seeds, disjoint from all v43, v89 and v90 roles. Each seed paired CFCMT and target only on the Los Angeles scenario and all three Jinan demands, giving 256 paired cells and 512 completed rollouts. The primary relative difference averaged scenarios within city, weighted Los Angeles and Jinan equally within seed and then averaged seeds; negative values favour CFCMT.

Target only achieved 149.09 s and CFCMT 155.14 s. The equal-seed relative difference was +4.254% (paired 95% interval $[-0.775, 13.211]$%), and only 29/64 seeds improved (one-sided exact sign-test probability 0.809). The city effects had opposite signs: CFCMT improved Jinan by 3.045% but regressed Los Angeles by 7.865%. The worst seed regressed by 267.56%, collision incidents were 25 for CFCMT and 15 for target only, and both had zero teleports. The predeclared mean, interval, both-city, worst-seed, improved-fraction, sign-test and collision gates therefore did not jointly pass, and the fresh source- contribution confirmation was rejected (Table 5).

Table 5: Closed-loop source-contribution ablation. Both policies use identical target adaptation groups, features, action executor and target scenarios; target only removes source-labelled samples from deployed prediction. Positive Δ denotes higher waiting for CFCMT. V90 is post-hoc; only V91 is a fresh-seed preregistered confirmation. Incident entries are CFCMT/target only.

Evidence	Seeds	Paired cells	Target-only wait	CFCMT wait	Δ wait	LA Δ	Jinan Δ	Improved	Incidents C/T
V90 post-hoc	2	8	149.04	149.17	+0.082%	+1.212%	-2.292%	7/8	0/1
V91 fresh (rejected)	64	256	149.09	155.14	+4.254%	+7.865%	-3.045%	29/64	25/15

2.6 Classical pressure remained the performance ceiling

MaxPressure and phase pressure achieved 141.74 and 143.62 s waiting, respectively, outperforming CFCMT by 7.43 and 5.55 s. The direction was the same in both external cities and the descriptive intervals excluded zero. Their system vehicle-hours and completion ratios were also better than CFCMT. This is a substantive boundary, not an implementation detail: the final CFCMT controller does not fall back to either rule. The pressure action used to construct a contrast is merely the zero point of a learned ranking, and any feasible phase can be selected at deployment.

All 56 rollouts completed with zero teleports. Across all policies there were 171 raw collision events consolidated into 13 unique junction incidents in six rollouts. CFCMT had zero incidents and never recorded more incidents than its paired comparator. Because incidents are simulator events and the matrix contains only eight CFCMT rollouts, this audit is a validity and failure report rather than a safety-certification claim.

2.7 A fresh-seed successor stress test did not close the pressure gap

After v43 was frozen, we separately tested whether a longer-horizon latent originator and a globally rate-limited executor could turn the pointwise ranking model into a controller that exceeded phase pressure. This successor track is not the anchored all-pair v43 controller and is not pooled with its two-city result. It uses a target-state-conditioned 120-s action originator, one permitted network-wide override at a time and a 120-s global switch cooldown on the Jinan target.

The legacy global executor appeared favourable on 54 disclosed adaptive seeds (-0.429% mean relative waiting difference; paired 95% interval $[-0.818, -0.033]\%$). Allowing unlimited cooldown stays instead regressed by 0.104%, and a one-stay budget produced an inconclusive -0.242% point estimate; both variants were rejected. We therefore froze the unchanged legacy executor before generating 64 new seeds disjoint from all adaptive seeds and from eight still-sealed prospective seeds.

On those 64 fresh seeds, the method–phase-pressure difference shrank to -0.043% (paired 95% interval $[-0.449, 0.370]\%$). The method improved 34/64 seeds, the one-sided exact sign-test probability was 0.354 and the worst seed regressed by 5.21%, just beyond the preregistered 5% limit. Teleports were 0/0 and collision incidents 1/1, but the bootstrap, sign-test and worst-seed gates failed. The joint confirmation was therefore rejected (Table 12). This untouched-seed result supports neither same-network superiority over phase pressure nor a cross-network claim for the successor executor.

2.8 Native reinforcement-learning results are protocol checks

Official RESCO and patched LibSignal experiments confirm that standard RL baselines can be executed in the project. The RESCO adapter completed 42 supported full-budget runs, and the LibSignal adapter completed 36 runs across DQN, PressLight, FRAP and MPLight. They use their native observations, rewards, action intervals, training loops and metrics, and therefore cannot be inserted into the 56-rollout table as if they were paired policy rows. Their results and upstream exclusions are reported in Tables 13 and 14; the same-protocol conclusion is based only on the external CFCMT matrix above.

3 Discussion

One supported result is about model structure: aligning the estimand and capacity with the decision improved a learned controller relative to a dense residual. CFCMT predicts a normalized ordering among phases from the same state, keeps a rigid local anchor and allows an all-pair correction only at a weight selected on complete held-out target seeds. This reduced external offline regret and closed-loop waiting relative to a same-information dense residual. Because the dense comparison also receives matched action contrasts, the gain cannot be explained by reference subtraction alone.

A distinct claim about transferred data did not survive fresh confirmation. The target-label-only comparator shares the target labels, feature semantics, model family and executor, but source-labelled samples have zero deployed weight. Source-plus-target CFCMT looked better in the seed-8171 offline action-regret comparison, yet on 64 new closed-loop seeds it helped Jinan and harmed Los Angeles. The equal-city mean favoured target only, the interval crossed zero and the collision count increased. Cross-network evaluation is therefore not evidence that source-city labels improve a target controller. The experiment instead identifies network-dependent negative transfer as a central failure mode of the present fitting rule.

The experiments also explain why the earlier full mechanism-world-model route was not retained as the final controller. MC-WM decomposed queue propagation, red accumulation, mobility, spillback and control cost. That decomposition is useful for diagnosis, but each auxiliary prediction introduces error before the model answers the narrower control question. Good next-state or mechanism loss does not guarantee the correct phase ordering, and flexible latent or graph paths can dominate the rigid signal under limited target support. The final method is therefore a reverse design from the successful rigid model: preserve declared parents and local mechanism features, then add only an antisymmetric ranking correction whose effect is measurable and bounded. This is a change in scientific hypothesis, not a residual patch to rescue a predetermined "full" architecture.

Pressure control remains the operational reference. MaxPressure and phase pressure were better than CFCMT in both external cities, with lower system vehicle-hours and slightly higher completion. Traffic-signal control has a strong local-rule ceiling because queue pressure directly reflects service imbalance. A learned method that beats fixed time or a dense neural baseline but not pressure control is not a replacement controller. CFCMT is better positioned as a transfer layer for settings where the decision depends on effects absent from a pressure score, or as a structured model used alongside rather than rhetorically above a rule. The discarded pressure-guard experiments in the supplement reinforce this point, but pressure is not hidden inside the final v43 policy.

The absence of same-protocol CrossLight, X-Light, MetaLight and GESA reproductions is also a substantive comparison gap, not merely a citation issue. Their published protocols use different target interaction budgets, state and action interfaces, simulators and training objectives. Reporting their native numbers beside the paired SUMO matrix would manufacture a ranking rather than estimate one. A broader benchmark should reimplement each method behind the same safe phase executor and expose three separate target budgets: zero labels, passive target transitions and matched counterfactual target actions.

Target information changes the interpretation. The target simulator is uncalibrated against field trajectories, which reduces calibration effort, but its action outcomes are labelled target data. The external result is thus a few-shot/full-budget offline-adaptation result with 165 and 911 groups, not a demonstration that a source-city policy can be deployed without target transitions. A genuine zero-shot arm remains scientifically useful as the left endpoint of an information curve, but it is not the basis of the present closed-loop claim. Conversely, field loop-detector or trajectory data could adapt passive state representations without supplying the counterfactual outcome of an unexecuted phase.

The target-only result also changes how the adaptation budget should be used. More source data is not monotonically beneficial after target labels are available. Three subsequent frozen gates tested whether source inclusion could be made a target-seed-blocked decision across the seven development city groups (Supplementary Table 9). Robust-history and out-of-fold action-agreement gates selected sources in six and seven cities, respectively, but each retained a worst-city regression above 0.02 and failed its target-only upper-bound criterion. A labeled five-mechanism compatibility gate removed these regressions only by selecting target only in all seven cities. Thus the additional experiments validate abstention as a safety fallback, not a source selector with positive transfer value.

Several limitations constrain generalization. First, the formal external unit count is two cities. Multiple Jinan demand files improve within-city coverage but do not create independent cities, so all intervals are explicitly descriptive. Second, the Los Angeles and Jinan scenarios are CityFlow-derived networks converted and semantically audited for SUMO; they are not calibrated city-wide digital twins. Third, training counterfactuals share the same SUMO physics family as evaluation. Cross-network transfer is real at the topology, demand and phase-program levels, but sim-to-real transfer is not tested. Fourth, the seven-policy closed-loop matrix uses two seeds, whereas the later source-contribution comparison uses 64 seeds but the same two target geometries. Both use a fixed 3,600-s horizon. Unfinished vehicles are included, yet longer incidents, non-recurrent demand and detector failure remain outside the experiment.

The sequential development record is another limitation and a strength. A diagnostic external seed exposed a one-city regression and motivated removal of an arbitrary adaptation cap and stricter complete-seed cross-fitting. That diagnostic seed was then permanently excluded, and a new seed authorized the final test. This is not equivalent to a single untouched preregistration; reporting the failed stages is necessary to expose researcher degrees of freedom. The final v43 cache, selectors, deployment models and closed-loop matrix are nevertheless hash-linked and frozen before their respective next stage.

A later execution-layer programme provides a stronger check on this boundary. Its favourable 54-seed adaptive estimate shrank almost to zero on 64 newly generated Jinan seeds and failed the preregistered bootstrap, sign-test and worst-seed criteria. Because that programme used a different 120-s latent originator and global cooldown, it neither rescues nor invalidates the v43 anchored all-pair comparison. It does show why adaptive search and independent confirmation must not be pooled, and why the present evidence cannot support learned-policy superiority over phase pressure.

The fresh target-only test is more directly damaging to a cross-city source benefit claim. Unlike the execution-layer programme, it holds the CFCMT model, target adaptation budget and executor fixed and changes only whether source labels enter deployed prediction. Its rejection cannot be dismissed as a different policy class. Offline regret and short-horizon action ranking did not reliably predict the closed-loop value of source data, particularly in the Los Angeles corridor where one seed produced a large congestion cascade.

The next decisive experiment is not a relaxed threshold or another unrestricted sweep on the same outcomes. It requires a new compatibility representation whose link to downstream action ranking is developed only on the current seven groups, then frozen and replicated on additional external cities and a calibrated digital twin or prospective signal intervention. Such a study could test whether the local pairwise advantage survives simulator mismatch and whether source mechanisms add value in objectives where pressure is incomplete, including multimodal priority, spillback risk, emissions or network-level coordination. Until then, the supported claim is narrower: causal parent restrictions and antisymmetric action ranking improve a simulation-assisted residual relative to dense modelling, while source-data benefit is network dependent and classical pressure remains the stronger controller on the reported targets.

4 Methods

4.1 Problem definition and information budget

Let c denote a city group, n a network, s_g a complete simulator state at matched action group g, and $\mathcal{A}_g$ the feasible green stages of the focal signal. A safe phase executor constructs this set from the original SUMO `tlLogic` and enforces minimum green, yellow and all-red transitions. The learner requests a feasible stage; it never writes an arbitrary signal string.

The final external setting is target offline adaptation. Available information comprises (i) a frozen source counterfactual bank, (ii) target network, demand and signal files, (iii) target counterfactual action groups from an uncalibrated SUMO model, and (iv) target state observed during deployment. No simulator parameter is calibrated against target field data, but item (iii) contains labelled target transitions. We therefore do not call the experiment zero-shot. Passive traffic observations without action outcomes would define a different, weaker information regime that is not used for the final policy result.

4.2 Matched counterfactual action groups

After a 60-s warm-up, the complete SUMO simulation and all signal-executor states are snapshotted. For every focal action $a \in \mathcal{A}_g$, the state is restored exactly, a is applied for one 10-s control interval, and the branch continues under phase pressure for five additional intervals. All other signals use phase pressure throughout the branch. The 60-s outcome is

$$Y_g(a) = \frac{1}{H}\sum_{\tau=1}^{H} \frac{N_{\text{active}}(\tau) + N_{\text{pending}}(\tau)}{L_n}, \qquad H = 60 \text{ s}, \tag{2}$$

where L_n is the number of controlled lanes. Pending vehicles are included so a phase is not rewarded for delaying insertion. The branch collector uses `libsumo`, checks replay determinism, fails on teleports, and censors an entire matched group when any branch has a junction collision. Group-level censoring prevents action-selective support from being retained.

A generalized pressure policy π_r, selected using source-only closed-loop cost, supplies one reference action $r_g = \pi_r(s_g)$. For the frozen external models its specification is `gp_d0_o0p02_s0`. Define

$$\sigma_g = \max\Big\{\max_a Y_g(a) - \min_a Y_g(a), \epsilon_g\Big\}, \tag{3}$$

$$z_g(a) = \frac{Y_g(a) - Y_g(r_g)}{\sigma_g}, \tag{4}$$

where ϵ_g is the numerical absolute/relative floor used by the implementation. The reference has $z_g(r_g) = 0$. Evaluation uses normalized action regret

$$\text{Reg}_g(\widehat{a}) = \frac{Y_g(\widehat{a}) - \min_{a\in\mathcal{A}_g} Y_g(a)}{\sigma_g}. \tag{5}$$

Assumption 1: action-common domain error. At one restored state, an absolute simulator or domain discrepancy can be written as

$$Y_g(a) = b_g + q_g(a) + e_g(a), \tag{6}$$

where b_g is independent of the candidate action. The remainder $e_g(a)$ may still be action dependent.

Proposition 1: exact cancellation. For any a, $Y_g(a) - Y_g(r_g) = q_g(a) - q_g(r_g) + e_g(a) - e_g(r_g)$; hence b_g cancels exactly. More generally, adding any state-specific constant to every candidate outcome leaves both Equation (4) and the action ordering unchanged.

This proposition motivates the estimand but does not remove action-dependent simulator error. Exact state restoration is therefore a protocol requirement, not merely a variance-reduction device.

4.3 Parent-restricted rigid anchor

The anchor $A_\theta(s_g, a)$ directly regresses $z_g(a)$ for non-reference candidates and assigns zero to the reference. Its 29 declared parents contain local total queue, vehicle count, speed and occupancy; candidate and reference green/red queue; downstream queue and occupancy; service and red pressure; switch and clearance indicators; and current-phase overlap. City identifiers, static target context and unrestricted graph embeddings are excluded from the direct predictor. These parents encode a modelling hypothesis about local action mechanisms; they were specified by design and were not discovered from field observational data.

The estimator is a `HistGradientBoostingRegressor` with 100 boosting iterations, at most 15 leaves, minimum leaf size 12 and L_2 regularization 10. Each city domain receives equal total weight, each matched action group receives equal weight within a domain, and negative versus non-negative candidate contrasts are reweighted to avoid the dominant sign determining the fit. Targets are normalized independently by σ_g. The 90th percentile absolute residual on source leave-one-domain predictions supplies the anchor uncertainty scale u_A.

4.4 Antisymmetric all-action correction

The correction model uses every unordered pair $\{a_i, a_j\} \subset \mathcal{A}_g$. Let x_g contain 23 action-common state parents and $\phi_g(a)$ contain 40 phase-local action parents. The training design and target are

$$d_{gij} = \left[x_g,\, \phi_g(a_i) - \phi_g(a_j)\right], \qquad t_{gij} = \frac{Y_g(a_i) - Y_g(a_j)}{\sigma_g}. \tag{7}$$

Every row (d_{gij}, t_{gij}) is augmented by its reverse $(d_{gji}, -t_{gij})$. At prediction, the regressor f_ψ is explicitly antisymmetrized,

$$\widehat{q}_{ij} = \tfrac{1}{2}\left[f_\psi(x_g, \phi_i - \phi_j) - f_\psi(x_g, \phi_j - \phi_i)\right], \tag{8}$$

and converted to an action score

$$P_\psi(s_g, a_i) = \frac{1}{K_g - 1}\sum_{j \neq i} \widehat{q}_{ij}, \qquad K_g = |\mathcal{A}_g|. \tag{9}$$

The group is finally shifted so the reference score is zero. Pair-cycle inconsistency augments the 90th-percentile residual uncertainty u_P.

Proposition 2: antisymmetry and ranking recovery. Equation (8) gives $\widehat{q}_{ji} = -\widehat{q}_{ij}$ by construction. If $\widehat{q}_{ij} = z_g(a_i) - z_g(a_j)$ for all pairs, then

$$P_\psi(s_g, a_i) = \frac{K_g}{K_g - 1}\left[z_g(a_i) - \overline{z}_g\right]. \tag{10}$$

The positive scale and subsequent reference shift preserve $\arg\min_a z_g(a)$. Thus an exact pairwise model recovers the optimal action without estimating an absolute state outcome.

The pairwise model uses the same gradient-boosting family as the anchor. Its state parents include local aggregate state, time, queue concentration and signal-graph neighbourhood summaries; its action parents include phase duration, queue, occupancy, downstream load, service pressure, transition and graph-action summaries. Context values are used only to calibrate support and uncertainty, not as direct prediction features.

4.5 Anchored interpolation and local cap

For one action group, the deployed score is

$$S_\alpha(s_g, a) = (1 - \alpha_g)A_\theta(s_g, a) + \alpha_g P_\psi(s_g, a), \qquad 0 \le \alpha_g \le 1. \tag{11}$$

Candidate selectors include constant weights $\alpha \in \{0, 0.1, \ldots, 1\}$ and local confidence-capped weights. For the local form, let

$$h_g = \max\{\mathrm{range}(A_g), \mathrm{median}(u_{A,g}), 10^{-6}\}, \tag{12}$$
$$D_g = \mathrm{range}(P_g - A_g)/h_g, \tag{13}$$
$$M_g = \max\{P_g(a_A) - P_g(a_P), 0\}/h_g, \tag{14}$$
$$U_g = \{u_P(a_A) + u_P(a_P)\}/h_g, \tag{15}$$

where $a_A = \arg\min A_g$ and $a_P = \arg\min P_g$. With cap c and confidence multiplier ζ,

$$\alpha_g = \min\{1, c/\max(D_g, 10^{-12})\}\mathbb{1}[M_g \ge \zeta U_g]. \tag{16}$$

Proposition 3: blend and decision-error bounds. For a group-constant α_g, if $\|A_g - z_g\|_\infty \le \varepsilon_A$ and $\|P_g - z_g\|_\infty \le \varepsilon_P$, then

$$\|S_\alpha - z_g\|_\infty \le (1 - \alpha_g)\varepsilon_A + \alpha_g \varepsilon_P. \tag{17}$$

For $\widehat{a} = \arg\min_a S_\alpha(a)$ and $a^* = \arg\min_a z_g(a)$,

$$z_g(\widehat{a}) - z_g(a^*) \le 2\|S_\alpha - z_g\|_\infty. \tag{18}$$

The first inequality follows from convexity of the sup norm; the second follows by adding and subtracting $S_\alpha(\widehat{a})$ and $S_\alpha(a^*)$ and using the minimizing property of $\widehat{a}$. These are deterministic conditional bounds, not guarantees that either fitted error is small in a new city.

Corollary 1: bounded local perturbation. Whenever the confidence indicator in Equation (16) passes,

$$\mathrm{range}(S_\alpha - A) = \alpha_g \,\mathrm{range}(P - A) \le c h_g. \tag{19}$$

When it fails, $S_\alpha = A$. The local rule therefore bounds score perturbation relative to the anchor, but it is not a queue-stability or safety guarantee.

4.6 Seed-blocked target selection and full refit

For each external city, all valid groups from adaptation seeds 5057 and 6067 are retained. In fold one, the source bank plus target seed 5057 fits both models and seed 6067 scores every candidate α; fold two reverses the target seeds. A candidate must be non-worse than the anchor in mean out-of-fold regret and may regress by at most 0.1 in either complete-seed fold. Among admissible candidates, the selector minimizes mean out-of-fold regret with deterministic tie breaking. Los Angeles selects $(c, \zeta) = (1, 0.25)$ and Jinan selects constant $\alpha = 0.9$.

After the choice is frozen, the model is fitted once on the source bank plus all target adaptation groups: 165 groups for Los Angeles and 911 for Jinan. The out-of-fold models are discarded. Method and learned baselines share the same target group identifiers and full-refit call. Model serialization is round-tripped before authorization. The seed-8171 evaluation cache is required to be absent at the joint freeze and the diagnostic-only seed 7079 is permanently excluded from v43 selection and confirmation.

4.7 Closed-loop controller and comparators

At each 10-s decision time, the runtime constructs every currently feasible phase, evaluates the frozen 60-s action score and requests the phase with minimum raw contrast. The safe executor handles physical transition timing. No uncertainty guard, pressure fallback or target closed-loop tuning is active in the final v43 policy. The pressure reference is recomputed only to create the zero coordinate required by the contrast model.

The same-protocol comparators are fixed time, MaxPressure, phase pressure, simulator-only contrast MPC, the rigid anchor alone and a dense residual. The dense baseline receives the same matched action-contrast data and refit budget but allows unrestricted feature/context mixing in a dense residual world model. It is labelled H2O$^+$-style because it instantiates the simulator-plus-learned-residual principle; it is not claimed to reproduce the original H2O$^+$ algorithm, optimizer or online training loop. This contrast-aligned baseline makes the comparison conservative: the CFCMT advantage cannot be attributed solely to subtracting the reference action.

Official RESCO and LibSignal controllers retain their upstream state, reward, timing and training definitions. They are native-protocol execution audits, not paired estimates of the CFCMT treatment contrast.

4.8 Target-label-only source-contribution test

The source contribution is isolated with the pre-existing `causal_target_only` model family. This comparator has the same causal action-advantage features, target domain and target adaptation groups as the corresponding baseline artifact. It sets prior group strength to zero and the maximum target weight to one. Both external artifacts had more than four target groups and selected target weights numerically equal to one. A source submodel is still fitted and retained for diagnostics, but its coefficient in the deployed mean, uncertainty and trust predictions is zero. The comparator is therefore target-label-only prediction with shared source-defined feature semantics, rather than an independently redesigned target controller.

The first deployment of this frozen comparator, v90, reused the eight v43 scenario–seed cells and was specified only after the original seven-policy outcomes were known. It is labelled post-hoc. The subsequent v91 protocol was written before any v91 outcome and did not refit either model. A deterministic generator produced 64 seeds disjoint from every seed-valued field in v43 and v89 and from the two v90 closed-loop seeds. Four scenarios and two policies gave 512 required rollout identities, all of which completed.

For policy π, city c, scenario set $\mathcal{Q}_c$ and seed s, we first compute

$$\bar{W}^{\pi}_{c,s} = |\mathcal{Q}_c|^{-1} \sum_{q \in \mathcal{Q}_c} W^{\pi}_{c,q,s}, \qquad \bar{W}^{\pi}_{s} = \frac{1}{2} \sum_{c} \bar{W}^{\pi}_{c,s}. \tag{20}$$

The paired seed statistic is

$$d_s = \frac{\bar{W}^{\mathrm{CFCMT}}_s - \bar{W}^{\mathrm{target}}_s}{\bar{W}^{\mathrm{target}}_s}, \tag{21}$$

where negative values favour CFCMT. The primary estimate is the mean of d_s over 64 seeds, with 20,000 paired-seed bootstrap replicates. The predeclared joint gate required a non-positive mean and interval upper bound, non-positive means in both cities, at most 5% worst-seed regression, at least 50% improved seeds, one-sided exact sign-test $p \leq 0.05$, no excess teleports or collision incidents, all identities and no failed-seed exclusion. Failure of any component rejects the source-contribution confirmation.

4.9 Separate post-freeze execution-layer stress test

The v86–v89 track is a distinct successor experiment on `jinan_3x4_real`. Its originator predicts a 120-s target-simulator cost from target-state and action features; the executor permits at most one network-wide override and requires at least 120 s between effective phase switches. The method was selected only from 54 disclosed adaptive seeds. Unlimited cooldown stays and a one-stay-per-cooldown variant were evaluated on the same adaptive partition and could not replace the legacy executor.

Before any v89 rollout, a deterministic generator selected 64 new seeds after excluding the 54 adaptive seeds and eight separately sealed v85 seeds. Each seed paired the frozen executor with phase pressure under the same target scenario. The preregistered joint rule required non-positive mean relative waiting difference, a non-positive paired-bootstrap 95% upper bound, no more than 5% worst-seed regression, at least 50% improved seeds, one-sided exact sign-test $p \leq 0.05$, at least 75% active seeds, no aggregate excess in teleports or collision incidents, closed phase accounting and all 128 rollout identities. Twenty thousand paired bootstrap replicates were fixed in the protocol. No failed seed was excluded, and v85 remained sealed. This track is reported as a stress test and is not part of v43 method selection.

4.10 Networks, splits and experimental execution

Development uses 18 networks in seven groups. Source model seeds are 2027 and 3037 and the excluded development evaluation seed is 4047. The successful development stage uses up to 100 target-style groups inside each held-out source-city exercise. External confirmation uses one Los Angeles network and one Jinan geometry with three demand files. Jinan scenarios are weighted equally before city aggregation, preventing that city from receiving triple the weight of Los Angeles.

Closed-loop evaluation runs four scenarios, two seeds and seven policies for 56 paired rollouts. Every rollout uses SUMO 1.22.0, `libsumo`, a 60-s warm-up and a 3,600-s measurement horizon. The experiment was parallelized across cluster workers, while numerical libraries were restricted to one thread per process to avoid nested oversubscription. Each rollout records its result and `tripinfo` hashes, source snapshot identity, model identity, SUMO version and metric population.

The fresh source-contribution confirmation uses the same four scenarios, simulator version, warm-up, horizon, action executor and metric implementation. Only the two frozen policies and 64 fresh seeds differ, producing 512 rollouts. Six process-isolated shards ran on six nodes with 22 workers per node. Full rollout records remain on the execution host; six complete shard summaries and the launch and audit manifests are materialized with the manuscript evidence.

4.11 Outcomes, aggregation and uncertainty

The preregistered closed-loop primary is mean vehicle waiting time. The implementation computes it over all departed vehicles, including unfinished `tripinfo` records whose waiting and duration are accrued to the fixed horizon. We additionally report completed-only waiting, system vehicle-hours, unfinished fraction, completion ratio, throughput, mean queue per lane, teleports, emergency stops and junction collisions. This metric bundle guards against a controller appearing good by withholding departures or by benefiting from right censoring.

Scenario-seed results are first averaged within scenario, scenarios are averaged equally within city and the two external cities are averaged equally. All policy contrasts remain paired on city, scenario and seed. Offline uncertainty uses 10,000 paired action-group bootstrap replicates. Closed-loop uncertainty uses 10,000 paired hierarchical replicates that sample city, then scenario, then seed. With only two independent target cities, these intervals describe sensitivity of the observed matrix; they are not estimates of a general population of cities. No multiplicity-adjusted claim or universal city-level significance is made.

The v91 source-contribution interval instead resamples the 64 paired seed statistics in Equation (21) 20,000 times. The exact sign test uses the number of seeds with $d_s < 0$. City-specific relative differences are secondary diagnostics because the independent city count remains two.

Raw SUMO collision events are consolidated by participant, lane and time into unique junction incidents. Teleports are a hard validity failure; collisions are retained as outcomes in closed loop and compared pairwise. A valid matrix requires all 56 authorized records, exact provenance identities, complete population accounting, readable `tripinfo` hashes and zero teleports. The v91 integrity audit analogously requires all 512 frozen identities and retains every seed regardless of performance or incident status.

4.12 Meaning of "causal"

CFCMT uses two causal design commitments: candidate actions are interventions from a common restored simulator state, and direct predictors are restricted to declared local parent sets. Proposition 1 is algebraic under an additive nuisance assumption, and Propositions 2–3 concern score construction. The study does not identify a causal graph from observational traffic, prove that the declared parents are invariant in the field, or estimate effects of signal interventions in Los Angeles or Jinan. "Causal" therefore describes the model and counterfactual simulation design, not field causal identification.

5 Conclusion

Cross-network traffic-signal transfer improved when the learned object was reduced from absolute dynamics to normalized, matched action contrasts. Anchoring a parent-restricted score and correcting it with an antisymmetric all-action model reduced seed-held-out action regret in development and in two external city-derived networks. The same frozen models improved closed-loop waiting time over a dense residual and produced favourable simulator-only and rigid-anchor point estimates, while remaining behind MaxPressure and phase pressure. Crucially, a 64-seed target-label-only comparison rejected a general source-contribution claim: source-plus-target fitting improved Jinan but harmed Los Angeles and was worse in the equal-city mean. CFCMT should therefore be interpreted as an auditable action-contrast architecture, not evidence that source data are uniformly beneficial. Establishing cross-city source benefit will require a precommitted source-selection rule, more independent target cities and either logged interventions, a validated digital twin or prospective field control; the present SUMO counterfactuals do not supply that evidence.

6 Data and Code Availability

The source benchmark combines public SUMO/RESCO scenarios (Lopez et al., 2018; Ault and Sharon, 2021) with public traffic-signal benchmark files used by LibSignal (Mei et al., 2024). The external Los Angeles input is the public LibSignal `LA_1x4` CityFlow roadnet and flow (Zhang et al., 2019), and the Jinan inputs are the public CoLight `Jinan/3_4` roadnet and three published flow files (Wei et al., 2019b). Conversion to SUMO, route conservation, signal-link identity, collision-free admission and exact source hashes are recorded in the v8 conversion manifest.

The manuscript source-data directory contains machine-readable CSV files for every new figure and table, together with a manifest linking the development audit, source and adaptation caches, deployment models, held-out result and closed-loop aggregate by SHA-256. It also contains the target-label-only source-contribution table generated from the v90 and v91 audits. The seven-city source-gate CSV and LaTeX table are regenerated directly from the frozen v145, v146 and v148 aggregate JSON files; all three decisions remain rejected. The 512 full v91 rollout records remain in the execution archive and are represented locally by six complete shard summaries. The current draft makes no claim of a public repository DOI. Before submission, the code snapshot, converted networks, configuration files, model artifacts, aggregate JSON, per-rollout results, `tripinfo` files and figure-generation scripts must be deposited in a persistent public archive subject to the licences of the upstream benchmark files; the DOI should replace this sentence in the accepted manuscript.

A Supplementary Methods and Results

A.1 Network inventory

Table 6: Source and external target network inventory parsed from the SUMO `net.xml` files. Jinan uses one geometry and three demand files.

Role	City group	Networks	TLS	Lanes	Scenarios
Source	atlanta	1	5	55	atlanta_1x5
Source	cologne	3	12	252	cologne1, cologne3, cologne8
Source	hangzhou	6	36	532	hangzhou_bc_tyc, hangzhou_kn_hz, hangzhou_qc_yn, hangzhou_sb_sx, hangzhou_4x4, hangzhou_4x4_hetero
Source	ingolstadt	3	32	2227	ingolstadt1, ingolstadt7, ingolstadt21
Source	new_york	1	196	2562	manhattan_28x7
Source	resco_synthetic	2	32	360	grid4x4, arterial4x4
Source	salt_lake_city	2	4	88	saltlake_400s_200w_q1_weekday_peak, saltlake_state_university_q1_weekday_peak
External target	los_angeles	1	4	157	la_1x4
External target	jinan	1	12	186	jinan_3x4 (3 demand files)

The external CityFlow files were converted to SUMO and revised through a semantic admission audit. Phase identities use final SUMO connection indices, permissive links remain distinct from protected movements, all routes are contiguous and demand is conserved. A 3,600-s admission run had no teleports or collisions. The admitted conversion tree has SHA-256 `4a2e17618c2ac584fc767562f3a7096c95d8f904f03ef6baa9cd5d34d282eadd`.

Table 7: Complete high-level decision trail for the pairwise method. Positive percentages denote lower normalized action regret than the rigid anchor. Seed 7079 is diagnostic evidence that informed v43 and is not reused as v43 confirmation.

Stage	Independent check	Result	Protocol consequence
v38 global pairwise	Salt Lake, 2 networks	−12.38% vs rigid; FAIL	reject global correction
v39 anchored selector	7-city development	−0.95% macro; FAIL	reject first anchor blend
v40 cross-fitted selector	7-city nested folds	selected anchor; FAIL	add seed-held-out evaluation
v41 seed-held-out	seed 4047, 7 cities	+9.84%, 6/7 improve; PASS	freeze for external diagnosis
v42 external diagnostic	seed 7079, 2 cities	+4.50% macro, 1/2 improve; FAIL	exclude seed; remove B100 cap
v43 external confirmation	seed 8171, 2 cities	+8.09%, 2/2 improve; PASS	authorize closed loop

A.2 Sequential development and confirmation

The v42 failure is retained because its city-macro improvement alone would have looked positive. It failed the predeclared requirement that both external cities improve: Los Angeles regressed by 0.00072 normalized regret. V43 then removed the arbitrary 100-group cap, used every valid adaptation group and cross-fitted by complete adaptation seed. The later seed 8171, rather than 7079, is the formal external confirmation.

A.3 City-resolved closed-loop outcomes

Table 8: External closed-loop outcomes before equal-city aggregation. Waiting is the all-departed 3,600-s metric in seconds. Incident entries are means over the two closed-loop seeds within that city and policy.

Policy	LA wait	Jinan wait	LA completion	Jinan completion	LA incidents	Jinan incidents
Fixed time	507.33	169.84	0.449	0.873	4.00	0.00
Dense residual	212.88	105.41	0.838	0.902	0.50	0.00
Simulator only	219.17	94.26	0.851	0.903	0.00	0.00
Rigid anchor	206.78	99.05	0.862	0.902	1.00	0.00
Anchored CFCMT	204.42	93.91	0.874	0.903	0.00	0.00
Phase pressure	197.51	89.73	0.890	0.905	0.50	0.00
MaxPressure	191.89	91.58	0.888	0.904	0.50	0.00

The CFCMT-versus-dense waiting improvements are 8.46 s in Los Angeles and 11.50 s in Jinan. CFCMT-versus-MaxPressure differences are −12.53 and −2.33 s, where negative values indicate that MaxPressure is better. The opposite directions cannot be hidden by equal-city aggregation.

A.4 Fresh source-contribution diagnostics

The target-label-only artifacts were created in the v43 baseline freeze, before the original closed-loop outcomes. Their selected target weights were within numerical precision of one in both cities. V90 added these artifacts to the eight original scenario–seed cells only after the main comparison and is therefore not a confirmation. V91 froze 64 unique seeds before execution, required 512 exact policy identities and completed all of them without seed exclusion.

The largest v91 failure occurred at seed 7417817 in Los Angeles. CFCMT waiting was 975.27 s with completion 0.270, compared with 194.74 s and completion 0.919 for target only; neither policy teleported or recorded a collision in that cell. This record is retained in the primary estimate. The failure was not solely an outlier artifact: across the 64 Los Angeles cells, the median relative difference was +2.75%, only 22 favoured CFCMT and 21 regressed by more than 5%. Across 192

Jinan scenario–seed cells, 179 favoured CFCMT. These diagnostics explain the opposite city means but do not change the preregistered seed-level analysis.

A.5 Seven-city source-gate development

Three later source gates were frozen and evaluated on the seven development city groups. The robust-history gate selected a source for six targets, and the action-agreement gate selected a source for all seven; both improved the equal-city mean but failed the target-only safety comparison because their one-sided upper bounds were positive and their largest city regressions exceeded 0.02 normalized regret. A stricter mechanism-compatibility gate used five B20-to-B5 target-label folds to score queue propagation, served movement, red accumulation, spillback and mobility. It admitted no source and returned exact target only in all seven cities (Table 9).

Table 9: Frozen seven-city source-gate development. Negative effects favour the selected gate. "Upper" is the one-sided 95% upper bound versus the architecture-matched target-only causal arm. Dense denotes the pooled H2O^{+}-style residual. These are development results, not external confirmation.

Version	Source gate	Source selected	Δ target mean	Upper 95%	Cities improved	Max. reg.	Δ dense mean
v145	Robust history	6/7	-0.0028	+0.0093	4/7	+0.0281	-0.0720
v146	OOF action agreement	7/7	-0.0016	+0.0095	3/7	+0.0262	-0.0708
v148	Mechanism compatibility	0/7	+0.0000	+0.0000	0/7	+0.0000	-0.0691

The progression exposes a safety–usefulness trade-off rather than a successful source selector. v145 and v146 retained source signal but allowed harmful city tails. v148 removed those tails only by abstaining everywhere. Its 7/7 advantage over the dense residual is therefore inherited from the rigid target-only causal model and cannot be attributed to source-city labels or to a full mechanism world model. No external source-gate confirmation was authorized after these failures.

A.6 Metric population and safety accounting

SUMO `tripinfo` was written with unfinished records at the fixed 3,600-s horizon. The primary waiting-time population includes all departed vehicles, with unfinished waiting accrued to the horizon. Completed-only waiting and completion ratios are secondary outcomes. System vehicle-hours integrate active plus pending vehicles and therefore include demand not yet inserted. Across the complete matrix, 171 raw collision events collapsed to 13 unique participant–lane–time incidents in six rollouts; CFCMT had zero. All starting and ending teleport counters were zero.

A.7 Discarded pressure-guard development track

Before the anchored pairwise method was developed, an exploratory controller placed a learned mechanism residual behind a MaxPressure fallback. On eight RESCO scenarios and 600-s rollouts its mean queue was 12.918, close to MaxPressure at 12.907. At 3,600 s, phase pressure was materially stronger (32.234 versus 43.511). These experiments motivated treating pressure as a hard performance ceiling, but they use a different controller, target set and metric from v43 and are excluded from the main confirmation.

Table 10: Exploratory 600-s pressure-guard stress test retained for development transparency. This is not the final anchored CFCMT controller.

Policy	Mean queue	Seed std	P90 queue	Trip time	Trip wait	Time loss	Throughput
MaxPressure	12.9068	0.1546	19.5279	87.5891	8.0605	24.5171	0.8267
CFCMT + pressure guard	12.9180	0.1919	19.5973	87.4863	8.0114	24.4127	0.8256
Phase pressure	13.2446	0.1816	19.3911	86.5371	7.4544	23.4628	0.8199
H2O+-style dense	17.2279	3.6084	28.9325	86.6585	8.5768	24.5488	0.7653
Target-static simulator	20.4023	0.1758	32.5538	97.9633	19.9032	35.3928	0.7429
Native actuated SUMO	30.1825	0.3536	47.6873	119.7813	33.0017	57.0805	0.7580
Fixed RESCO program	42.6732	0.2709	69.6852	142.3116	54.6333	79.8759	0.7127

Table 11: Exploratory 3,600-s pressure-guard boundary check. Phase pressure is the strongest policy in this discarded protocol.

Policy	Mean queue	Seed std	P90 queue	Trip time	Trip wait	Time loss	Throughput
Phase pressure	32.2343	0.9901	62.1800	104.2291	20.8528	38.5434	0.9492
MaxPressure	42.7872	0.7314	71.6277	106.9182	20.5521	40.7331	0.9448
CFCMT + pressure guard	43.5106	0.5699	74.2496	108.6326	22.0400	42.4485	0.9448
Target-static simulator	44.8973	1.2335	65.4094	258.2547	169.8900	188.9974	0.8754
H2O+-style dense	49.5058	3.9510	74.9880	221.4173	131.1706	150.3275	0.8827
Native actuated SUMO	58.3468	1.1370	99.4055	164.5754	66.1326	98.1604	0.9714
Fixed RESCO program	84.6088	0.6458	133.0312	221.2093	113.3988	154.3776	0.9385

A.8 Post-freeze fresh-seed execution-layer stress test

The v86–v89 successor track uses a target-state-conditioned 120-s action originator and global execution cooldown rather than the v43 anchored all-pair controller. Negative deltas favour the learned executor relative to phase pressure. The first three rows reuse the same 54 disclosed adaptive seeds and are not independent replications. Only v89 uses 64 newly generated seeds; its joint gate was frozen before those rollouts and failed. The eight v85 prospective seeds were not opened.

Table 12: Post-freeze Jinan execution-layer stress test. Intervals are paired simulator-seed bootstrap intervals. Development rows are shown to expose the adaptive-to-confirmatory shrinkage and are not pooled with v89.

Stage	Evidence	Seeds	Mean Δ	95% CI	Improved	Decision
v86 legacy global	adaptive screen	54	-0.429%	[-0.818, -0.033]%	68.5%	development only
v87 unbounded stay	adaptive ablation	54	0.104%	[-0.295, 0.519]%	46.3%	variant rejected
v88 one-stay budget	adaptive ablation	54	-0.242%	[-0.674, 0.195]%	55.6%	variant rejected
v89 frozen legacy	fresh confirmation	64	-0.043%	[-0.449, 0.370]%	53.1%	confirmation rejected

A.9 Native-protocol reinforcement-learning checks

A.10 Frozen artifact chain

The publication artifacts are regenerated with

```
python3 paper_tsc/figures/build_tsc_external_confirmation.py
python3 paper_tsc/figures/build_tsc_causal_transfer_schematic.py
python3 paper_tsc/figures/build_tsc_postfreeze_stress_test.py
python3 paper_tsc/figures/build_tsc_source_gate_development.py
```

Table 13: Official RESCO full-budget RL results under the upstream native protocol. Unsupported upstream scenario–algorithm pairs are not silently patched into new methods. These rows are not same-protocol comparisons with CFCMT.

Algorithm	Scenario	Runs	Time loss	Duration	Waiting	Elapsed (s)
IDQN	arterial4x4	3	1505.536	501.305	436.920	17179.205
MPLight	arterial4x4	3	1056.797	357.454	238.745	13576.024
IDQN	cologne1	3	612.737	164.905	138.950	812.441
MPLight	cologne1	3	189.393	122.751	68.843	6985.784
IDQN	cologne3	3	86.610	92.313	43.268	2351.500
MPLight	cologne3	3	1011.239	543.511	515.329	10132.753
IDQN	cologne8	3	24.760	89.379	8.354	5946.643
IDQN	grid4x4	3	47.752	158.215	25.205	11120.623
MPLight	grid4x4	3	44.393	156.112	16.598	10392.985
IDQN	ingolstadt1	3	18.097	37.191	5.490	911.332
MPLight	ingolstadt1	3	401.895	85.906	63.192	7888.143
IDQN	ingolstadt21	3	256.588	353.111	176.405	13141.256
IDQN	ingolstadt7	3	51.869	78.827	17.357	5313.646
MPLight	ingolstadt7	3	284.391	111.560	57.463	8755.139

Table 14: Patched LibSignal full-budget execution audit under native LibSignal states, rewards, timing and metrics.

Agent	Network	Runs	Travel time	Queue	Delay	Throughput	Elapsed (s)
dqn	sumo1x1	3	38.945	3.240	0.283	1998.667	175.967
frap	sumo1x1	3	41.923	24.461	0.460	1664.667	439.620
mplight	sumo1x1	3	40.546	4.373	0.319	1995.000	371.569
presslight	sumo1x1	3	39.748	3.442	0.285	1998.667	175.835
dqn	sumo1x3	3	97.090	6.531	0.244	2487.333	425.384
frap	sumo1x3	3	118.712	11.672	0.398	2505.333	1583.053
mplight	sumo1x3	3	66.502	1.876	0.243	2814.333	647.664
presslight	sumo1x3	3	63.188	1.164	0.188	2816.000	393.414
dqn	sumo4x4	3	140.610	0.603	0.041	1465.000	2140.773
frap	sumo4x4	3	140.477	0.612	0.042	1464.000	7188.185
mplight	sumo4x4	3	140.469	0.607	0.042	1464.333	1017.297
presslight	sumo4x4	3	144.279	0.711	0.048	1463.333	2080.043

These commands form the figure and source-table submission contract. The unreferenced v16-era builder and its outputs are retained only as development history and must not be substituted for Figures 1–3. The first command refuses to write figures or tables if the development, external offline, joint-freeze or closed-loop validity gate is not satisfied. The third command separately requires all v86–v89 evidence-integrity audits to pass and the preregistered v89 scientific decision to remain rejected. The fourth command requires the v145, v146 and v148 development decisions to remain rejected and does not promote them to external confirmation.

Table 15: SHA-256 chain for the final external result. The paper artifact builder verifies all result gates before regenerating this table.

Artifact	SHA-256
development audit	5bcae9467cdb4cc157676e94078d204702ebdca6d51007ffca4bc85d13951066
source cache	14c6dfad87999e55fa86fa942ac0ca2334129f4d1662b3c31a6e4fc7aa74c242
target adaptation cache	beb6a968c43dd58ceac4df6d9da8dfca47fc1f0869dfda0dc151a3e9cae58d77
joint method freeze	471e8c34eb7be96411c812c7fe961d5c07f1857e3fb50d2c8d7edd59fc23c596
held-out seed-8171 result	20d682b48bca0f8821ecf98e54b04c94af87d530add51cc1d92d9a176f74cf60
closed-loop aggregate	be36c73c5835e632231140fdbdabc4bba422f54b503160c0179efc902d4534e0
post-freeze target-only ablation	238dbdc27ceb7ce016b9c995b5472cf1ad71a76632b6609b0df66408a1e26a7e
fresh target-only confirmation	3a0ee874afeff6df96a90fb4aed4e42f79705469f0d2a72da514932385990f67
Los Angeles deployment model	f26ee8e5829e248ef411f77ad3e19ed0d63fd52cdfdae6d8291a9a3aeaaa7aff
Jinan deployment model	a34970f1e4f023079650866097359ec3b2509e17787685249a21363cc7b03741
closed-loop source tree	ad555ff72a11a62cc88c8bbbe2ab2f5df1db8de920af03e2f61d99aaba226a7d

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